The Riemann Hypothesis as the positivity of a measure

The Riemann Hypothesis can be stated as a positivity condition. The reciprocal Xi function is the Laplace transform of a generalised gamma convolution exactly when every zero lies on the critical line, and then its Thorin measure is the discrete measure on the squared ordinates. This volume is the theory of that condition.

It gives every representation of the Xi function the two volumes use, the classes of infinitely divisible laws in which they live, from generalised gamma convolutions and their extended form to hyperbolic complete monotonicity, and the reciprocal Xi function at every basepoint. It follows the Thorin condition through the reciprocal zeta function, the critical strip, Dirichlet L functions and the Dedekind eta function, and through hitting times, extreme values and ferromagnetic random variables.

The last chapters read the Thorin measure as a mixture of Cauchy laws, prove a strip version of the Lee and Yang theorem for spins of Xi type, and assemble a route to the Riemann Hypothesis in six equivalent forms, each proved at the line Re s = 1, with curves over finite fields as the benchmark where the same route is a proof. Every statement is marked proved, computed, certified, conditional, heuristic or open.

the same measure at eight resolutions, from $\varepsilon = 1$ (bottom) to $\varepsilon = 0.005$ (top)

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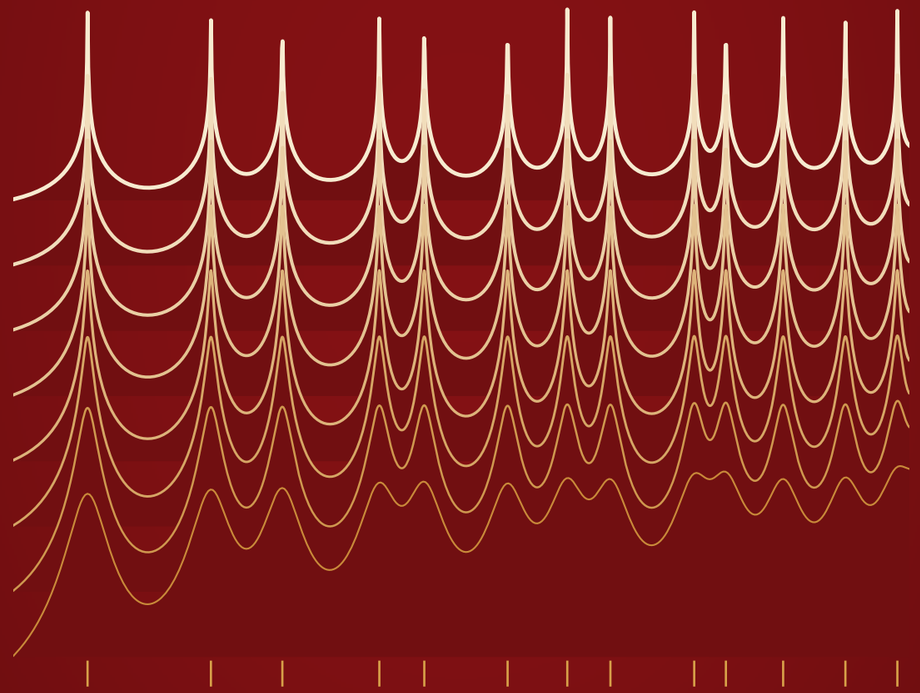
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Computing the Riemann Hypothesis

Volume I

Representations of Xi, Thorin Measures and the Reciprocal Xi Function



the Thorin density of $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ sharpening onto the zeros

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